

To Smooth or Not? When Label Smoothing Meets Noisy Labels

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1 / 24

Why we care: Machine learning with imperfect labels is transforming the world

- We scale modern ML by training on massive, cheaply labeled datasets (web, crowdsourcing).
- We routinely face imperfect labels in vision, NLP, and recommendation systems.
- We observe that noisy labels hurt generalization, slow training, and can miscalibrate confidence.

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2 / 24

Our background: Label Smoothing (LS) as a ubiquitous regularizer

- We use LS [Szegedy et al., 2016] to replace one-hot labels with a mix of hard labels and the uniform distribution.
- We find benefits on clean data: better generalization and calibration across tasks.
- We ask: how should we regularize labels when they are already noisy?

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3 / 24

Our question: When does LS help or hurt under noisy labels?

- We see reports that LS can improve robustness at low/moderate noise.
- We also see that at high noise rates, LS can oversmooth and wash out useful signal.
- We aim to give a principled rule to choose between LS, vanilla CE, or alternatives.

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4 / 24

Our challenge: We need to shape confidence under noise

- We recognize that noisy labels reduce the effective confidence signal during learning.
- We note that extra smoothing can further depress confidence, leading to underfitting.
- We seek a mechanism that boosts confidence for correct patterns while suppressing it for wrong ones.

Our idea and contributions

- We unify positive LS and Negative Label Smoothing (NLS) via Generalized Label Smoothing (GLS).
- We show a theory-driven phase transition: LS at low noise; NLS at high noise.
- We derive a closed-form optimal smoothing r_{opt} and connect NLS to robust losses.
- We provide extensive evidence and a practical recipe (CE warm-up $\rightarrow$ NLS).

Our method: Generalized Label Smoothing (GLS)

- We define GLS: $y^{\text{GLS},r} = (1-r) \cdot y + (r/K) \cdot \mathbf{1}$, extended to $r \in (-\infty, 1]$.
- We recover LS when $r > 0$ (probability mass toward uniform) and vanilla when $r = 0$.
- We recover NLS when $r < 0$ (negative weighting; labels sum to 1 though entries may be negative).

How we choose LS vs NLS

- We choose r based on the noise rate e and the clean-data preference r^* .
- Our decision map (figure) yields LS when $r_{\text{opt}} > 0$, vanilla when $r_{\text{opt}} = 0$, and NLS when $r_{\text{opt}} < 0$.
- Reading guide: above the boundary is the NLS region; below is LS; the switch occurs where $r_{\text{opt}} = 0$.
- We observe on UCI datasets the same phase transition: as noise grows, the best r turns negative, matching r_{opt} predictions.

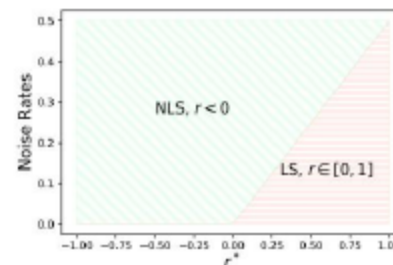


Figure: Our decision between NLS and LS given e and r^* .

Our lens: Risk equivalence under noisy labels (Theorem 3.2)

- We show that minimizing a loss with smoothed noisy labels equals the clean target risk plus bias terms.
- We analyze how the bias terms directly modulate model confidence under symmetric/asymmetric noise.
- We use this lens to explain why LS reduces and NLS increases confidence in targeted ways.

How we connect to robust learning methods

- We show loss correction (forward/backward) [Patrini et al., 2017] is recoverable as NLS with specific negative r .
- We link complementary labels [Ishida et al., 2017; Ishida et al., 2019] to the limit $r \rightarrow -\infty$ (an NLS view).
- We interpret Peer Loss [Liu and Tao, 2020] as an NLS-regularized objective under certain priors.
- We unify why these methods work: all shape confidence under noise.

Our phase transition formula for symmetric noise

- Under symmetric noise rate e , we obtain $r_{\text{opt}} = \frac{r^* - 2e}{1 - 2e}$.
- We see three regimes: $e < r^*/2 \Rightarrow$ LS ($r_{\text{opt}} > 0$); $e = r^*/2 \Rightarrow$ Vanilla; $e > r^*/2 \Rightarrow$ NLS ($r_{\text{opt}} < 0$).
- We note the multiclass extension follows similarly with ϵ .

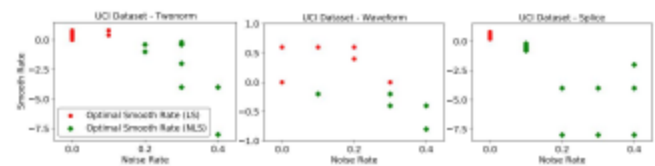


Figure: Optimal smooth rates on UCI datasets vs. label noise (ties possible).

How we train: a practical recipe for NLS

- We warm up with CE to learn clean patterns, then switch to NLS.
- From ablations, direct NLS can collapse early (overconfidence on noise); warm-up stabilizes training and unlocks NLS gains.
- We find negative r robust at high noise; moderately negative values (e.g., $r \approx -0.4$ to -4) work well; larger $|r|$ may require longer warm-up.

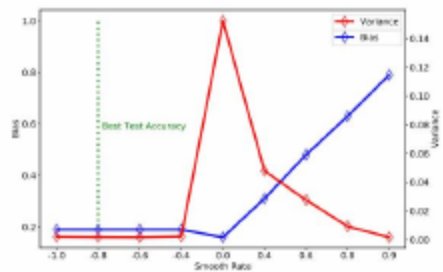


Figure: Bias and variance of LS/Vanilla/NLS models on clean CIFAR-10.

- We evaluate on synthetic 2D, UCI (Image, Waveform, Heart, Twonorm, Banana), CIFAR-10/100, AGNews, CIFAR-N, Clothing1M.
- We vary noise types (symmetric/asymmetric) and noise rates.
- We measure test accuracy and analyze phase transitions and confidence calibration.
- Our baseline suite includes CE, LS [Szegedy et al., 2016], NLS (ours), loss correction [Patrini et al., 2017], SCE [Wang et al., 2019], APL [Ma et al., 2020], Peer Loss [Liu and Tao, 2020], ELR [Liu et al., 2020], Bootstrap [Reed et al., 2015], AUM [Pleiss et al., 2020], and an f -divergence method [Zhang and Sabuncu, 2018].

- Label Smoothing (LS): [Szegedy et al., 2016]
- Forward/Backward Loss Correction: [Patrini et al., 2017]
- Complementary Labels: [Ishida et al., 2017; Ishida et al., 2019]
- Peer Loss: [Liu and Tao, 2020]
- Symmetric Cross-Entropy (SCE): [Wang et al., 2019]
- Active Passive Loss (APL): [Ma et al., 2020]
- Early-Learning Regularization (ELR): [Liu et al., 2020]
- Bootstrap: [Reed et al., 2015]
- AUM: [Pleiss et al., 2020]
- f -divergence method: [Zhang and Sabuncu, 2018]

What we see under noise: confidence becomes the bottleneck

- We see that with high noise, LS becomes uncertain over large regions (undershoots).
- We see NLS sharpen confidence on correct regions and suppress it on wrong ones.
- We find this matches the confidence-bias interpretation from our theory.
- What to look for: left=NLS retains confident regions on the true manifold; right=LS diffuses confidence and blurs boundaries; vanilla is in between.

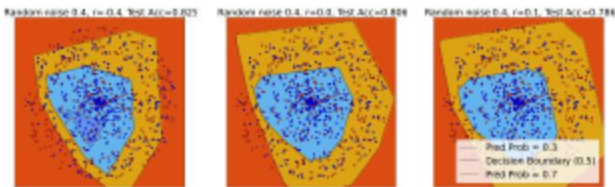


Figure: Confidence maps on synthetic Type 1 data at $\epsilon=0.4$ (left: NLS, middle: Vanilla, right: LS).

What we observe on CIFAR-10/100: NLS wins at higher noise

- We find that on CIFAR-10 (asymmetric), NLS is competitive and yields significant gains at $\epsilon = 0.2$.
- On CIFAR-100 (symmetric), at $\epsilon = 0.6$ NLS beats the best LS by ~ 5 points (very small p -values).
- We also see the same LS $\rightarrow$ NLS switch on AGNews as noise rises (supports generality beyond vision).

Table: Table 4. Test accuracy (mean $\pm$ std) comparisons on asymmetric noisy CIFAR-10, symmetric CIFAR-100 datasets

Smooth Rate	CIFAR-10 Asymmetric		CIFAR-100 Symmetric	
	$\epsilon = 0.2$	$\epsilon = 0.3$	$\epsilon = 0.4$	$\epsilon = 0.6$
$r = 0.8$	90.45 $\pm$ 0.06	87.83 $\pm$ 0.13	54.04 $\pm$ 0.93	39.50 $\pm$ 0.18
$r = 0.6$	90.41 $\pm$ 0.09	87.83 $\pm$ 0.13	52.72 $\pm$ 0.15	40.49 $\pm$ 0.07
$r = 0.4$	90.49 $\pm$ 0.10	87.90 $\pm$ 0.13	54.26 $\pm$ 0.07	41.57 $\pm$ 0.05
$r = 0.0$	88.32 $\pm$ 0.24	86.27 $\pm$ 0.32	48.03 $\pm$ 0.29	38.11 $\pm$ 0.14
$r = -0.4$	87.27 $\pm$ 1.83	88.33 $\pm$ 0.06	56.87 $\pm$ 0.08	43.70 $\pm$ 0.16

What we observe in the wild: Clothing1M and CIFAR-N

- We find that NLS beats LS and strong baselines on Clothing1M (+0.8% over LS).
- We observe that across CIFAR-N splits, NLS achieves best or tied-best results.
- We highlight that GLS is simple, drop-in, and effective beyond synthetic noise.

Note: F-div in the table refers to: [Zhang and Sabuncu, 2018].

Table: Table 6. Performance comparisons on Clothing 1M and CIFAR-N: results of baselines are obtained through the public leader-board.

Method	Clothing 1M	CIFAR-10N Aggre	CIFAR-10N Rand1	CIFAR-10N Worse	CIFAR-100N Fine
CE	68.94	87.77	85.02	77.69	55.50
BLC	69.13	88.13	87.14	77.61	57.14
FLC	69.84	88.24	86.88	79.79	57.01
PL	72.60	90.75	89.06	82.53	57.59
F-div	73.09	91.64	89.70	82.53	57.10
LS (best)	73.44	91.57	89.80	82.76	55.84
NLS (best)	74.24	91.97	90.29	82.99	58.59

How we affect representations: larger margins with NLS

- We observe that NLS increases inter-class separation in pre-logits.
- We see sharper confidence translate to cleaner class clusters.
- We take this as evidence that NLS restores discriminative structure under noise.
- What to look for: tighter, well-separated clusters with NLS in pre-logit space indicate restored structure.

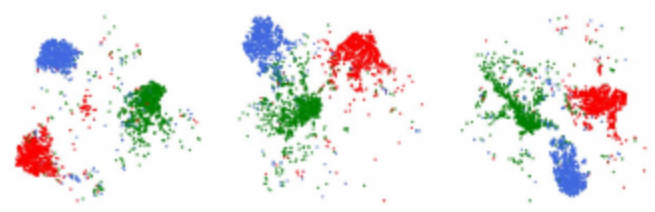


Figure: Pre-logit embeddings on CIFAR-10 (left: NLS, middle: Vanilla, right: LS).

How we shape confidence on CIFAR-10

- We see NLS concentrate high confidence on correct predictions.
- We see it reduce confidence on wrong predictions, improving reliability.
- We note that LS distributes confidence more moderately, which can underfit under noise.
- What to look for: we see higher confidence for correct and lower for wrong with NLS, widening separation compared to LS.

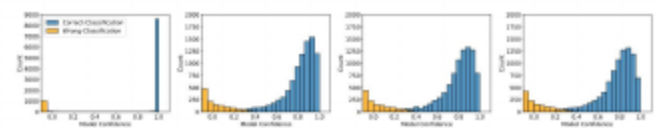


Figure: Confidence distributions on CIFAR-10 test (NLS, Vanilla, LS; trained with symmetric 0.2 label noise).

Our limitations and practical considerations

- Estimating noise rate or r^* is nontrivial; very large negative r without warm-up can destabilize training.
- We note that state-of-the-art pipelines (e.g., ELR [Liu et al., 2020]) may still outperform GLS alone in some settings.
- We treat the r schedule as a hyperparameter search in practice.

- We unify LS and NLS in GLS and explain their effects via confidence shaping.
- We recommend a phase-aware strategy: use LS at low noise; switch to NLS as noise grows.
- We advocate a simple, effective recipe: CE warm-up $\rightarrow$ NLS with moderately negative r .
- We clarify why several robust methods work (as NLS instances).

- We aim for automatic r scheduling via noise estimation or validation signals.
- We plan to combine NLS with advanced noisy-label pipelines (selection, semi-supervised).
- We will extend to detection, segmentation, and language modeling with varied label quality.
- We encourage rethinking label regularization beyond the probability simplex.

- Thank you for your attention!
- We welcome questions and feedback.

- We thank collaborators, reviewers, and open-source contributors.
- We are grateful to dataset maintainers and the research community.